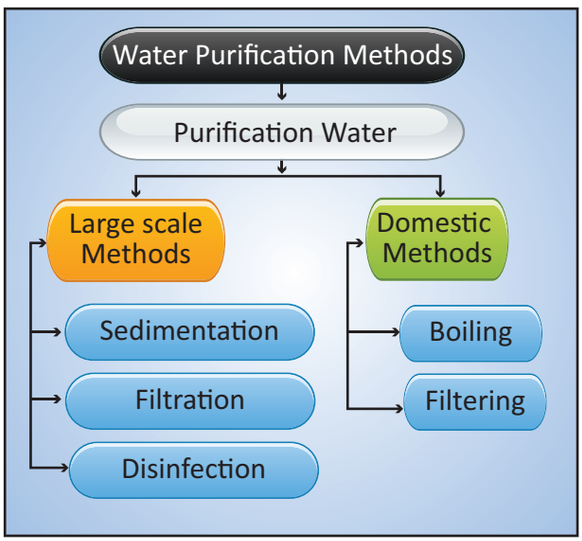
8.3 Water Safety

Safe drinking water is most important for consumption. It is necessary to protect drinking water from contamination for public health, economic and environmental reasons.

8.3.1 Importance of Safe Drinking Water

Safe drinking water is the basic necessity of a community. The availability and quality of water determines the quality of life. Thus water is a finite resource that has limits and boundaries to its availability and suitability for use.

Water purification methods help to ensure availability of safe drinking water. **Water purification** is the process of removing undesirable chemicals, biological contaminants, suspended solids and gases from water. Purifying water may reduce the concentration of particular matter including suspended particles, parasites, bacteria, algae, viruses, fungi, as well as reducing the concentration of a range of dissolved and particular matter.



8.3.2 Water Purification Methods

Industrial Methods (Large Scale Methods)

1. **Sedimentation:** Water taken from deep borewells and similar underground sources may be almost free from suspended impurities. But river,tank and lake waters carry a large amount of suspended matter, causing the water to be turbid and coloured. Storage in reservoirs makes these impurities settle to the bottom. The settlement and removal of impurities is effective by adding chemical agents to the water such as alum. The chemical hydrolysis in water forms flaky, flocculent precipitates in which the suspended particles including a large number of bacteria are trapped and carried to the bottom by sedimentation.
2. **Filtration:** Sedimentation may not remove all suspended impurities in water. Therefore water is further purified by filtration. For the filtration of water for human consumption two types of filters are used. They are Slow Sand Filters and Rapid Sand Filters used in industries.
3. **Disinfection:** The filtered water is disinfected. The most common disinfectant used is chlorine. Chlorine is a strong oxidant that rapidly kills many harmful micro-organisms.Other possible pathogens include viruses, bacteria, including *Salmonella, Cholera,Campylobacter* and *Shigella*. Chlorine is added to water in the form of solution of chlorine gas drawn from steel

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